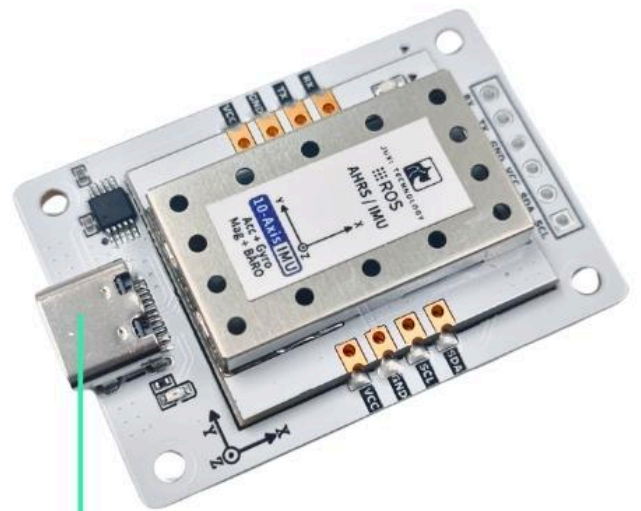


Raspberry Pi 5

1. Connect the device

This tutorial takes the Raspberry Pi 5 motherboard and the official 64-bit version of the mirroring as an example.

Connect the IMU attitude sensor to the USB port of the main controller via a Type-C cable.



USB

2. Check device status

View device ID

代码块

```
1  lsusb
```

View Device Number

代码块

```
1  ls -l /dev/ttyU*
```



A terminal window titled 'pi@pi: ~' with a menu bar containing '文件(F)', '编辑(E)', '标签(T)', and '帮助(H)'. The terminal shows the following commands and output:

```
pi@pi:~ $ lsusb
Bus 004 Device 001: ID 1d6b:0003 Linux Foundation 3.0 root hub
Bus 003 Device 001: ID 1d6b:0002 Linux Foundation 2.0 root hub
Bus 002 Device 001: ID 1d6b:0003 Linux Foundation 3.0 root hub
Bus 001 Device 002: ID 1a86:7523 QinHeng Electronics CH340 serial converter
Bus 001 Device 001: ID 1d6b:0002 Linux Foundation 2.0 root hub
pi@pi:~ $ ls -l /dev/ttyU*
crwxrwxrwx 1 root dialout 188, 0  2月  3日 12:04 /dev/ttyUSB0
pi@pi:~ $
```

Set Port Mapping

代码块

```
1  # 防止插拔后端口变更, 请设置端口映射
2  sudo gedit /etc/udev/rules.d/99-serial-imu.rules
3
4  # 如出现没有gedit命令相关内容, 先下载安装
5  sudo apt install gedit
6
7  # 填写映射内容
8  KERNEL=="ttyUSB*", ATTRS{idVendor}=="1a86", ATTRS{idProduct}=="7523",
   MODE=="0777", SYMLINK+="imu-serial"
9
10 # 参数说明:
11 `--mode`: 通信模式, 可选值为`serial`(串口)或`i2c`
12 `--port`: 串口名(如`/dev/ttyUSB0`)或I2C端口号(如`7`)
13 `--rate`: 数据打印频率(Hz), 默认10Hz
14 `--debug`: 启用调试模式, 显示详细信息
15
16 # 保存退出, 运行命令使规则生效
17 sudo udevadm trigger
18 sudo service udev reload
19 sudo service udev restart
20
21 # 验证
22 ll /dev/imu-serial
23
24 # 输出示例:
25 lrwxrwxrwx 1 root root 7 1月 22 10:00 /dev/imu-serial -> ttyUSB0
```

3. Install the driver library

3.1 Install the Python libraries required for the code

代码块

```
1 sudo apt update
2 sudo apt install -y python3-serial
3 sudo apt install -y python3-smbus2
```

3.2 Transfer Files



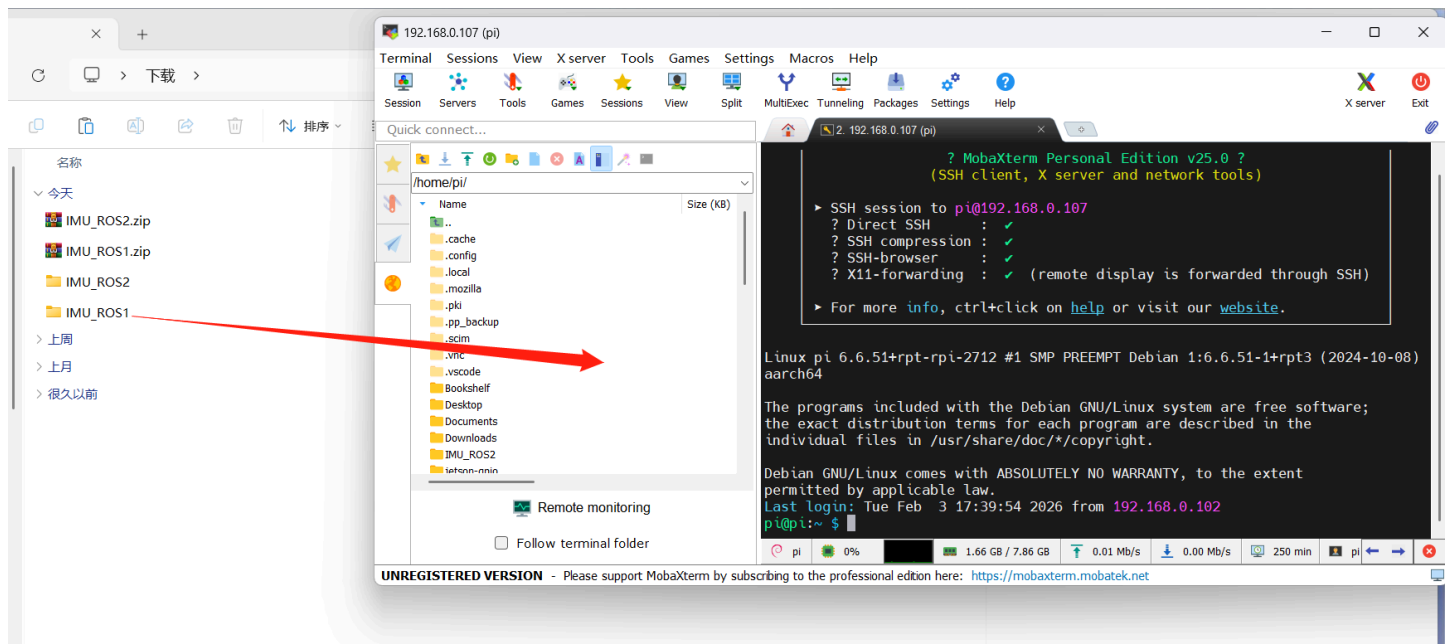
IMU_ROS2.zip

357.96KB



Friends who are not yet familiar with using MobaXterm to transfer files, please refer to the following webpage for detailed installation and operation methods of MobaXterm: [国文件远程传输](#)

Drag the decompressed files onto Raspberry Pi 5 via MobaXterm software.



4. View IMU data

Enter the ~/IMU_Library directory and run the IMU_Serial_Library.py file

代码块

```
1 cd ~/IMU_ROS2/IMU_Library
2
3 # 运行 IMU 数据打印文件
```

```
4 python3 -m IMU_Library.IMU_Serial_Library
```

```
pi@pi: ~/IMU_ROS2/IMU_Library
文件(F) 编辑(E) 标签(T) 帮助(H)
pi@pi:~/IMU_ROS2/IMU_Library $ python3 -m IMU_Library.IMU_Serial_Library
<frozen runpy>:128: RuntimeWarning: 'IMU_Library.IMU_Serial_Library' found in sys.modules after import of pac
kage 'IMU_Library', but prior to execution of 'IMU_Library.IMU_Serial_Library'; this may result in unpredicta
ble behaviour
当前系统: Linux
IMU串口已打开! 波特率: 115200
成功打开串口: /dev/ttyUSB0
-----数据接收线程已启动-----
固件版本: V1.1.2
数据报告频率已设置为: 10 Hz
加速度计: [0.07617419965208899, 0.5464033936582537, -0.8520767845698416]
陀螺仪: [0.0, 0.0, 0.0]
磁力计: [-2.0020142216254158, -13.818781090731528, 43.214209418012025]
姿态角: [152.14785957673848, 1.347570709726438, 150.88054529728865]
四元数: [0.0710824579000473, 0.24011078476905823, 0.9388784766197205, 0.22897391021251678]
气压计: [0.01, 18.64, 100970.48438, 100970.5625]

加速度计: [0.0771507919553209, 0.5454268013550219, -0.8515884884182257]
陀螺仪: [0.0, 0.0, 0.0]
磁力计: [-1.782280953398236, -13.818781090731528, 43.31186864833521]
姿态角: [152.04096711625218, 1.2746645234904281, 150.23225106217575]
四元数: [0.07200367003679276, 0.24548566341400146, 0.9372636675834656, 0.2296011596918106]
气压计: [0.12, 18.63, 100969.10156, 100970.5625]

加速度计: [0.07666249580370495, 0.5464033936582537, -0.8525650807214575]
陀螺仪: [0.0, 0.0, 0.0]
磁力计: [-2.0020142216254158, -13.892025513473921, 43.140964995269634]
姿态角: [151.97810205514574, 1.2666764038145104, 149.58059637399197]
四元数: [0.07337616384029388, 0.25077229738235474, 0.9357274770736694, 0.22971898317337036]
气压计: [0.16, 18.63, 100968.60156, 100970.5625]
```

Note: The above is the data reading for a 10-axis IMU. The 6-axis has no Magnetometer and Barometer data, and the 9-axis has no Barometer data.

5. IMU Calibration

Enter the ~/IMU_Library directory and run the imu_calibration_tool.py file

代码块

```
1 cd ~/IMU_Library
2
3 # 运行 IMU 校准代码文件 --串口通讯校准
4 # 执行所有校准 (整体、磁力计、温度)
5 python3 -m IMU_Library.imu_calibration_tool --mode serial --port /dev/imu-
  serial
6
7 # 仅整体校准
8 python3 -m IMU_Library.imu_calibration_tool --mode serial --port /dev/imu-
  serial --calibrate imu
9
10 # 仅磁力计校准
```

```
11 python3 -m IMU_Library.imu_calibration_tool --mode serial --port /dev/imu-  
    serial --calibrate mag  
12  
13 # 仅温度校准  
14 python3 -m IMU_Library.imu_calibration_tool --mode serial --port /dev/imu-  
    serial --calibrate temp
```

6. Precautions

If the device ID can be found but the device number cannot, you can refer to the following command to install the ch34x driver

代码块

```
1 sudo apt remove brltty  
2 git clone https://github.com/clhchan/CH341SER.git  
3 cd CH341SER  
4 make -j6  
5 sudo make install  
6 sudo modprobe ch34x
```